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To the Editors of *Forestry: An International Journal of Forest Research*

Dear Editors,

please consider the enclosed manuscript, “How much of the stem must a terrestrial laser scanner see? Diameter error as a function of circumferential coverage, and what it means for scan design”, for publication as an Original Article in *Forestry*.

When stem diameters are derived from terrestrial laser scanning (TLS) point clouds, the processing algorithms accept or reject a diameter depending on how much of the stem circumference was scanned. 3DFin, published in *Forestry* in 2024, accepts a diameter only if at least 9 of 16 angular sectors (202.5°) contain points. How much coverage is actually needed, and whether one threshold suits all fitting methods, has not been quantified. The manuscript does this with three complementary approaches:

- a Monte Carlo experiment with six circle and cylinder estimators under realistic perturbations (non-circular stems, outliers, registration offsets);
- HELIOS++ simulations of three stand densities and a stand with branches at breast height, with the visible arc computed geometrically and independently of point density (26 712 stem × scan-configuration cases plus 9 360 with branches);
- two open real datasets (SegmentedForests, LAUT) and 3DFin itself run on twelve real plots with simulated reduced coverage.

The main findings are directly relevant to inventory practice: the coverage threshold depends on the estimator and on how well stem points are separated from branches; least-squares fits fail at any coverage when branch points remain, whereas an arc-constrained RANSAC does not; 3DFin’s rule keeps accepted diameters accurate (88 % within 2 cm) but leaves about half of the stems without a diameter even with four to six spread scan positions; and spreading scan positions matters more than adding them.

All code and result tables are archived on Zenodo; all third-party data are openly licensed.

The manuscript is original, has not been published and is not under consideration elsewhere. The author declares no competing interests and received no specific funding.

Yours sincerely,

Angel Naya